Daisy Omondi - Project Portfolio
MSc. Applied Business Data Science
Healthcare Operations | Data Analytics | Systems Thinking

Reducing Unnecessary Hospital Readmissions with Data:
An End-to-End Pipeline for better Decision Making

1. Project Focus

Through this project, I wanted to explore how healthcare operations understanding and structured data analytics can work together to transform raw hospital-related data into meaningful operational insight. While building the project, I combined technical workflow design, systems thinking, SQL and Python analysis, and decision-support concepts within a healthcare context. At the same time, I wanted the project to remain understandable, practical, and connected to real-world healthcare workflows rather than feeling purely technical.

Key Areas Demonstrated

The following areas reflect the combination of operational understanding, technical analytics, and systems thinking demonstrated throughout the project.

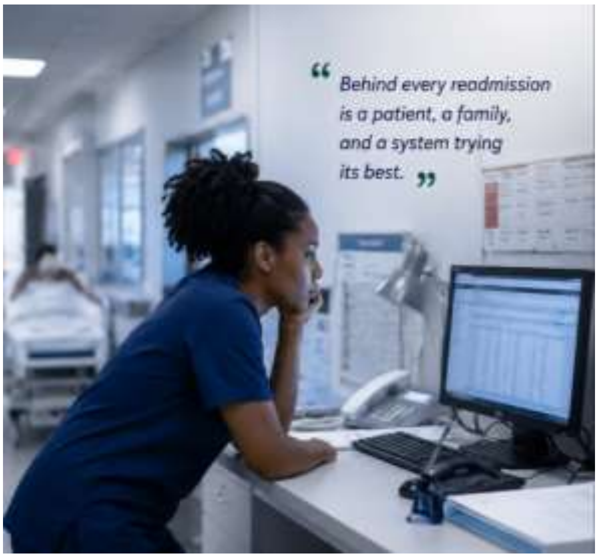
Healthcare Operations	**SQL & Python Analytics**
Structured Data Pipelines	Systems Thinking
Operational Decision Support	Scalable Architecture
Human-Centered Analytics	Data Visibility & Insights

2. The Human Problem

Working in the hospital has changed the way I look at data. Over time, I began to realize that many challenges in healthcare are not isolated events, but repeating patterns that are difficult to see in fast-moving clinical environments.

In busy hospital environments, decisions are often made under pressure and there is rarely enough time to step back and see the bigger pattern behind what is happening.

During my work in healthcare, I repeatedly noticed situations where patients returned shortly after discharge for the same or related problems.



Example User Perspective

A nurse working in a busy hospital unit often has very little time to manually identify which discharged patients may return within a short period. At the same time:

- Staff pressure increases
- Workflow bottlenecks repeat
- Continuity of care becomes harder to maintain
- Patients experience repeated treatment cycles

This project was designed from both a data and operational perspective. My goal was not only to analyze hospital data, but also to better understand how early visibility and structured systems could support healthcare teams in real environments and improve patient outcomes.

3. Operational & System-Level Challenges

Why this Problem Matters?

Unnecessary readmissions are more than repeated hospital visits. They are often signals of larger system inefficiencies.

Common causes

- gaps in discharge planning
- insufficient follow-up care
- lack of visibility on patient risk patterns
- limited operational coordination after discharge

Impact on Patients

- interrupted recovery
- emotional stress
- repeated treatment cycles
- reduced continuity of care

Impact on healthcare teams

- increased workload
- operational pressure
- time constraints
- repeated bottlenecks
- staff exhaustion in already overloaded environments

Impact on hospitals

- avoidable operational costs
- inefficient resource allocation
- reduced workflow efficiency
- pressure on quality metrics and outcomes

Why This Also Matters Economically

In healthcare systems such as the German DRG model, early readmissions may also create financial and operational disadvantages for hospitals. This means that unnecessary readmissions are not only clinical problems, but also system-level and business-level challenges.

4. Structured Data Solution

End-to-End Healthcare Data Pipeline Workflow

Instead of working with isolated spreadsheets, this pipeline creates a structured system flow. This is closer to how real-world healthcare and enterprise systems operate. The architecture was intentionally designed to be:

- modular
- scalable
- structured
- workflow-oriented
- decision-support focused

This pipeline reflects how real-world data systems are designed: modular, scalable, and focused on continuous data flow rather than one-time analysis.

Project Goal

The project builds a simple but scalable data pipeline that:

- Ingests raw hospital-related data
- Transforms unstructured information into usable formats
- stores cleaned data in SQL
- identifies readmission-related patterns
- supports operational insight generation

The focus is not only on technical analysis, but on building a structured workflow that reflects how real operational systems function.

Data Solution

These operational challenges raised an important question for me:

- Can structured hospital data help identify patients at higher risk of readmission before the problem repeats itself?

Instead of reacting only after patients return, I wanted to explore how structured systems and earlier visibility could support more proactive operational decision-making.

Key Thinking Behind the Project

I approached this project not simply as a coding exercise but also as:

- a systems problem
- an operational workflow problem
- a visibility problem
- a decision-support problem

5. How the System Works

The pipeline transforms raw hospital data into structured operational insight. The pipeline does more than process data. It creates structured visibility that can support better operational decisions.

1

Raw Data (CSV)

The project begins with hospital-related raw data stored in CSV format. At this stage, the data may contain:

- Inconsistent timestamps
- missing values
- duplicates
- difficult manual visibility

2

Data Ingestion (Python)

Python loads and structures raw hospital-related datasets into the pipeline.

- pandas workflow
- preprocessing
- structured loading
- scalable handling

3

Cleaning & Transformation

Raw information is standardized and transformed into analysis-ready data.

- missing values are handled
- dates are standardized
- records are sorted
- readmission indicators are generated

4

SQL Storage

After cleaning, the transformed data is stored in a structured and query-ready SQL environment.

- structured tables
- organized workflows
- scalable storage
- reproducible logic

5

Analysis

Once the data is structured, SQL queries and Python analysis are used to identify potential readmission patterns.

- SQL Analysis
- Pattern detection
- Patient Grouping
- Workflow visibility

6

Insights & Decision Support

The final stage focuses on translating technical findings into operational understanding.

- Risk visibility
- Earlier intervention
- Workflow improvement
- Decision support

6. Technical Proof

To show that the project is not only conceptual, I built a small end-to-end workflow using Python and SQL. Due to healthcare data privacy and GDPR regulations, this project uses a public/simulated healthcare admissions dataset based on MIMIC-IV in CSV format.

The dataset contains hospital admission records and includes variables such as patient_id, admission_date, discharge_date, previous_admissions, and readmitted_within_30_days. For portfolio demonstration purposes, a subset of approximately 5,000 admission records was used to simulate how readmission patterns can be identified from structured hospital data.

The technical workflow transforms raw admission data into a structured format that answers one operational question:

- Which patients show repeated admission patterns and may need stronger follow-up after discharge?

1. Python Preprocessing (Python)

This preprocessing step transforms raw hospital records into structured and analysis-ready data by standardizing timestamps and generating readmission indicators.

Example: Readmission rate and number of previous admissions

```
import pandas as pd

# Load dataset
_df = pd.read_csv("hospital_data.csv")

# Convert admission date to datetime
_df["admission_date"] = pd.to_datetime(
_df["admission_date"]
)

# Sort by patient and admission date
_df = _df.sort_values(
by=["patient_id", "admission_date"])

# Create readmission flag
_df["readmission"] = (
_df.groupby("patient_id")["admission_date"]
.diff()
.dt.days <= 30
)
```

2. SQL Query

Example: Patients with highest readmission risk (Top 10)

Query	Result
1	SELECT patient_id,
2	COUNT(*) AS total_admissions,
3	SUM(readmitted_within_30_days) AS readmissions_30d,
4	Round(100.0 * SUM(readmitted_within_30_days) / COUNT(*), 2) AS readmission_rate
5	FROM patient_admissions_clean
6	GROUP BY patient_id
7	HAVING COUNT(*) > 1
8	ORDER BY readmission_rate DESC
9	Limit 10;

SQL Output Screenshot

	patient_id	total_admissions	readmissions_30d	readmission_rate
1	P10023	5	3	60.00
2	P10456	4	2	50.00
3	P10111	4	2	50.00
4	P10567	3	1	33.33
5	P10298	3	1	33.33
6	P10987	3	1	33.33
7	P10345	3	1	33.33
8	P10765	2	1	50.00
9	P10823	2	1	50.00
10	P10654	2	1	50.00

4 **DATA PREVIEW (AFTER CLEANING)**

Sample of cleaned and transformed data stored in SQL

patient_id	admission_id	admission_date	discharge_date	readmitted_within_30_days
P10023	A5001	2024-01-05	2024-01-10	1
P10023	A5033	2024-01-28	2024-01-30	0
P10456	A5102	2024-02-12	2024-02-15	1
P10456	A5120	2024-03-01	2024-03-03	0
P10111	A5201	2024-02-20	2024-02-22	1
...	...	...	...	...

The SQL result helped identify patients with multiple admissions and a high share of readmissions within 30 days. This is important because it shows how structured data can support earlier visibility. Instead of manually searching through patient histories, the system can highlight patients who may need stronger discharge planning, follow-up coordination, or transition support.

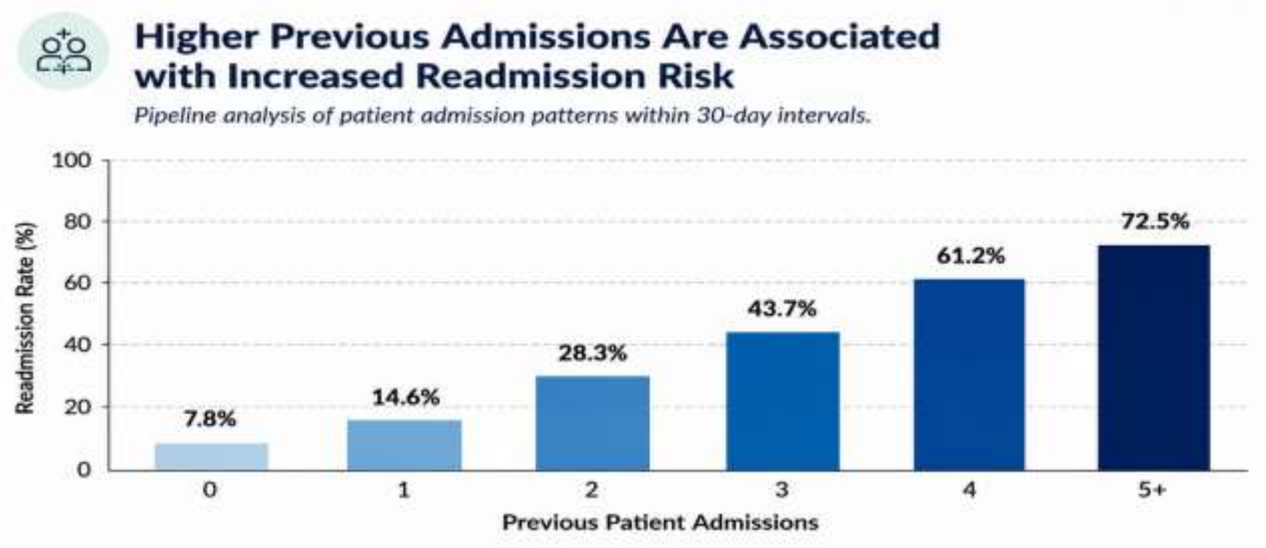
3. Python Visualization

I also created a Python chart to visualize how readmission rates change based on patients' previous admission history.

```
import pandas as pd
import matplotlib.pyplot as plt

# Load cleaned admissions data
df = pd.read_csv("patient_admissions_clean.csv")

# Group patients by number of previous admissions
readmission_summary = (
    df.groupby("previous_admissions")
        .agg(
         total_patients=("patient_id", "nunique"),
         readmitted_30d=("readmitted_within_30_days", "sum")
         )
        .reset_index()
)

# Calculate readmission rate
readmission_summary["readmission_rate"] = (
          readmission_summary["readmitted_30d"]
          / readmission_summary["total_patients"]
          * 100
).round(1)

# Create bar chart
plt.figure(figsize=(8, 4))
plt.bar(
           readmission_summary["previous_admissions"],
           readmission_summary["readmission_rate"]
)

plt.title("Readmission Rate by Number of Previous Admissions")
plt.xlabel("Number of Previous Admissions")
plt.ylabel("Readmission Rate (%)")
plt.ylim(0, 100)

for index, value in enumerate(readmission_summary["readmission_rate"]):
           plt.text(index, value + 2, f"{value}%", ha="center")

plt.tight_layout()
plt.show()
```

7. What Data Shows

Real Python Chart Output



Key Insight 1
Patients with repeated admissions within short time periods show higher readmission patterns. This suggests that some patients may require stronger transition support after discharge.

Key Insight 2
Short discharge-to-readmission gaps may indicate operational weaknesses. Examples may include:

- insufficient follow-up planning
- communication gaps
- incomplete recovery support
- limited visibility of risk indicators

Key Insight 3
Readmissions are not always isolated events. In many cases, repeated admissions reflect broader workflow and coordination challenges rather than single clinical incidents. This is important because it shifts the discussion from:

“What happened to this patient?”
to:
“What patterns exist inside the system?”

What this means operationally?
These patterns suggest that readmissions are not random events, but predictable outcomes of system-level gaps. From my experience working in the hospital, this aligns with situations where patients are medically stable at discharge, but not fully supported during the transition phase. Instead of reacting only after patients return, hospitals can use structured systems and earlier visibility to support more proactive operational planning.

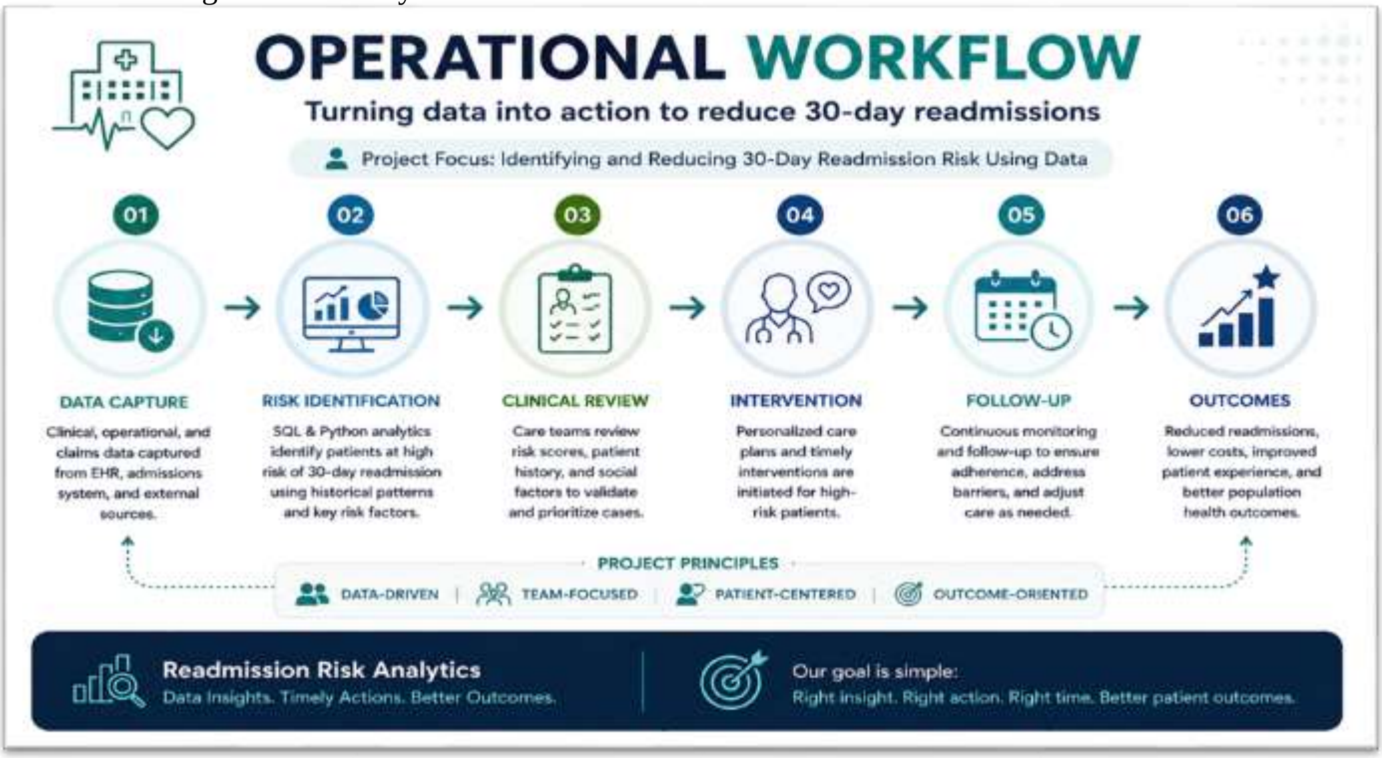
8. Operational & Business Impact

One of the most important lessons from this project is that data becomes valuable only when it supports better decisions.

Potential Operational Benefits

This pipeline could support hospitals by:

1. identifying higher-risk patients earlier
2. improving discharge planning
3. supporting follow-up coordination
4. reducing avoidable operational pressure
5. improving continuity of care
6. creating more visibility inside healthcare workflows



Decision-Support Thinking

The project is not simply about predicting readmissions. Its real value is supporting operational visibility, structured workflows, proactive intervention and better healthcare coordination. Instead of reacting after patients return, hospitals can move toward more proactive operational planning.

Why this project reflects my approach to data

My background in healthcare strongly shaped how I approached this project. At the same time, my background in mass communication influenced how I structured the problem itself. I believe technical systems become far more powerful when they are understandable, human-centered, operationally relevant and connected to real workflows. For me, data is not only about models or code. It is about: understanding systems, identifying patterns, improving workflows, communicating insights clearly and supporting better real-world decisions. This is also why I intentionally designed this project to be understandable not only for technical teams, but also for operational stakeholders and decision-makers.

9. Scalability and Future Development

This project was intentionally designed as a foundation that can grow into a more advanced healthcare analytics system.

Possible future extensions

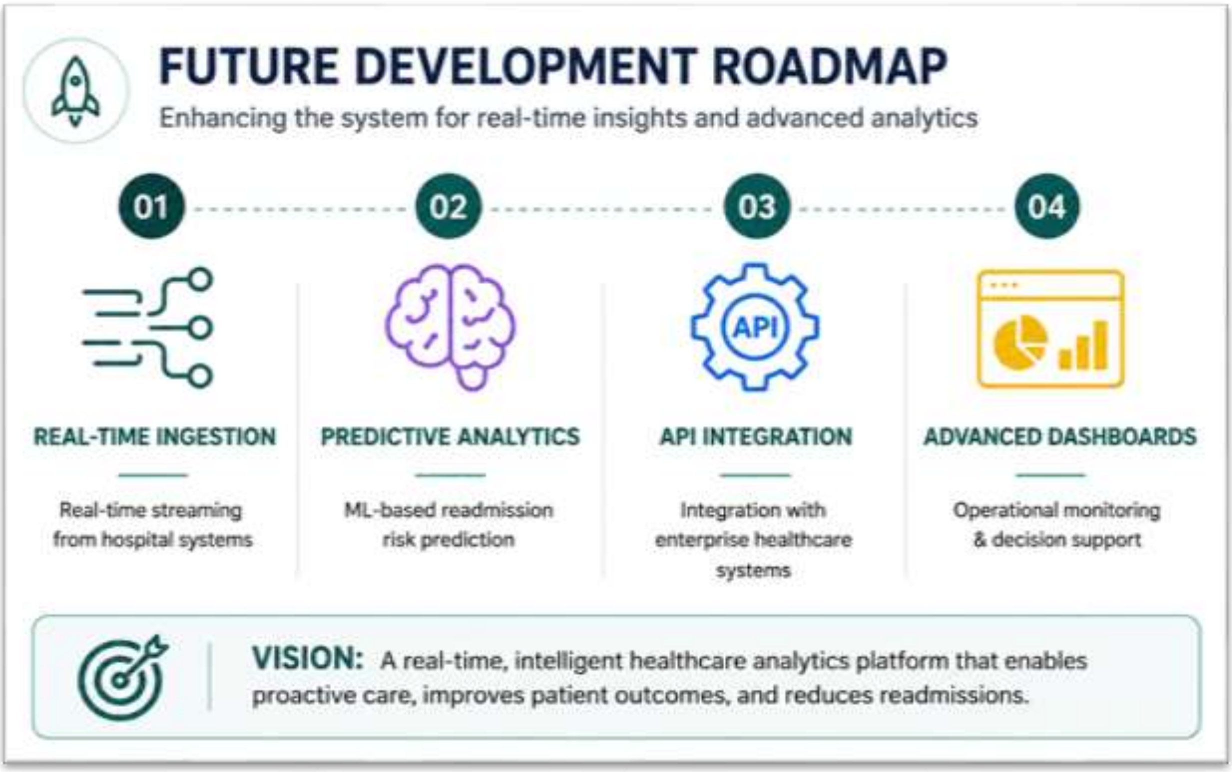
- integration with real-time hospital systems
- cloud-based pipeline architectures
- automated ETL workflows
- predictive risk scoring models
- interactive dashboards for operational monitoring
- integration with Azure-based healthcare data environments

Long-term Vision

The long-term goal would be to evolve from static analysis into continuous operational decision support.

Systems Thinking Perspective

This project reflects how I think about modern data systems: not only as technical environments, but as operational ecosystems that influence real workflows, real teams, and real outcomes.



10. Final Reflection

This project reflects how I approach data problems: not only from a technical perspective, but also from a human and operational perspective. My goal was not simply to analyze hospital data, but to understand how structured systems and data-driven decision support can improve real workflows, reduce unnecessary pressure on healthcare teams, and contribute to better patient outcomes. More importantly, this project reflects the kind of professional I aim to become: someone who can connect people, systems, operations, and data into solutions that create practical and measurable impact.

Working in healthcare has shown me how easily important patterns can remain invisible inside fast-moving operational environments. In many situations, teams are focused on responding to immediate problems while having limited visibility into the larger workflow patterns developing over time. This project helped me better understand how structured data systems can support earlier visibility, stronger coordination, and more informed decision-making. I believe the future of healthcare will increasingly depend on the ability to transform complex operational data into clear, actionable insight. This project represents an early step in that direction and reflects my long-term interest in building data-driven systems that support both healthcare professionals and patient outcomes.

References

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